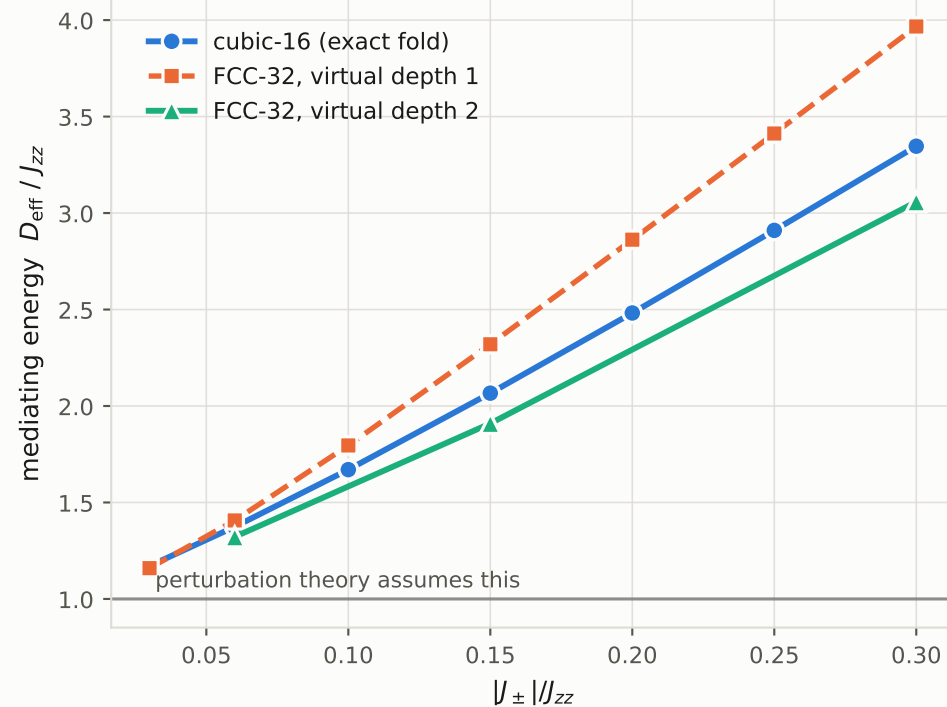
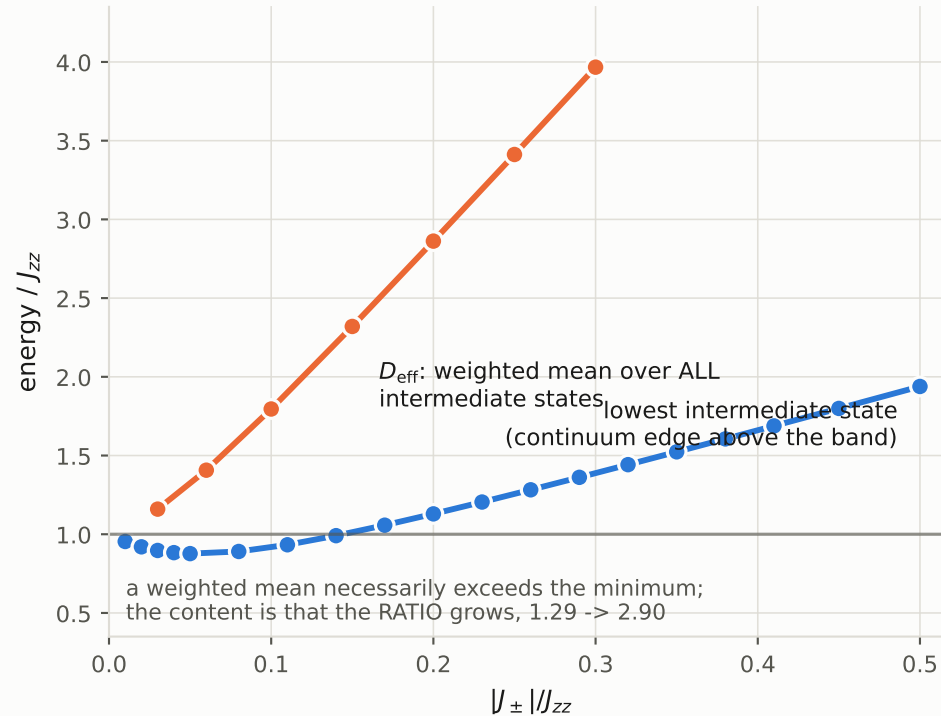


What one ring-exchange move costs, and why it matters

(a) Doubling the crystal barely moves it



(b) Both scales grow; the denominator outruns the gap



(c) $\text{Ce}_2\text{Hf}_2\text{O}_7$: where the photon actually is

